MCQ-9

Topic: Machine Learning – Clustering, PCA, Ensemble Learning & Boosting

Student Name: _____ **Date:** _____ **Score:** _____

Instructions: Select the correct option by marking the checkbox ☐ next to it.

1. Two centroids are $C_1 = (2, 3)$ and $C_2 = (8, 7)$. A point $P = (5, 4)$ must be assigned using Euclidean distance. Which cluster will P belong to?

   ☐ C_1, because $d(P, C_1) = \sqrt{10}$ and $d(P, C_2) = \sqrt{18}$
   ☐ C_2, because $d(P, C_1) = \sqrt{18}$ and $d(P, C_2) = \sqrt{10}$
   ☐ C_1, because $d(P, C_1) = \sqrt{13}$ and $d(P, C_2) = \sqrt{13}$
   ☐ C_2, because $d(P, C_1) = \sqrt{10}$ and $d(P, C_2) = \sqrt{10}$

2. Which combination best describes an important characteristic of XGBoost compared with a basic gradient-boosted decision-tree implementation?

   ☐ It eliminates gradient-based optimization by training trees independently.
   ☐ It uses regularization and a second-order optimization formulation, along with engineering optimizations such as efficient tree construction.
   ☐ It is simply Random Forest with gradient descent added.
   ☐ It requires neural networks as its base learners instead of decision trees.

3. A cluster contains $(2, 4)$, $(4, 6)$, $(6, 8)$, $(8, 10)$. After the assignment step, these points belong to the same cluster. What is the new centroid?

   ☐ $(5, 7)$
   ☐ $(5, 8)$
   ☐ $(4, 7)$
   ☐ $(6, 8)$

4. A dataset has covariance-matrix eigenvalues $\lambda_1 = 5$, $\lambda_2 = 3$, $\lambda_3 = 2$. If PCA retains only the first two principal components, what percentage of the total variance is retained?

   ☐ 50%
   ☐ 70%
   ☐ 80%
   ☐ 90%

5. Consider 1-D points $1, 2, 3, 7, 8, 9$. DBSCAN uses $\varepsilon = 1.1$ and MinPts $= 3$, counting the point itself. Which statement is correct?

   ☐ Points $1, 2, 3, 7, 8, 9$ are all core points.
   ☐ Points 2 and 8 are core points; the remaining points are border points.
   ☐ Points $1, 2, 3, 7, 8, 9$ are all border points because no point has 3 neighbors.
   ☐ Points 2 and 8 are noise because they have only two ε-neighbors.

6. Three binary classifiers predict: $M_1 =$ Class 0, $M_2 =$ Class 1, and $M_3 =$ Class 1. A hard voting ensemble uses equal weights. What is the final prediction?

   - ☐ Class 0, because the first classifier has the lowest index
   - ☐ Class 0, because the classifiers must agree completely
   - ☐ Class 1, because it receives the majority vote
   - ☐ Class 1 only if soft voting is used

7. A regression model initially predicts $\hat{y} = [4, 7, 10, 12]$, while $y = [6, 5, 11, 15]$. Gradient Boosting uses $r_i = y_i - \hat{y}_i$. What residual vector should the next weak learner fit?

   - ☐ $[2, -2, 1, 3]$
   - ☐ $[-2, 2, -1, -3]$
   - ☐ $[10, 12, 21, 27]$
   - ☐ $[4, 7, 10, 12]$

8. Which statement about PCA is most accurate?

   - ☐ PCA selects original features with the highest individual variance.
   - ☐ PCA constructs orthogonal linear combinations of features that successively maximize variance.
   - ☐ PCA guarantees that every transformed feature has higher variance than every original feature.
   - ☐ PCA is supervised because it maximizes variance with respect to class labels.

9. Which statement most accurately distinguishes AdaBoost from Gradient Boosting?

   - ☐ AdaBoost can only perform regression, while Gradient Boosting can only perform classification.
   - ☐ AdaBoost explicitly reweights training examples based on previous errors, while Gradient Boosting fits new learners to the negative gradient/residual of a specified loss.
   - ☐ Both algorithms always use exactly the same loss function and update equations.
   - ☐ Gradient Boosting changes sample weights directly, while AdaBoost always fits residuals from squared-error loss.

10. Consider $A = (0, 0)$, $B = (0, 1)$, $C = (1, 0)$, and $D = (1, 1)$. DBSCAN uses $\varepsilon = 1.01$ and MinPts $= 3$, including the point itself. How many clusters are formed?

    - ☐ 0
    - ☐ 1
    - ☐ 2
    - ☐ 4

11. Suppose a dataset contains Age measured from 0–100 and Income measured from 0–$10,000,000$. PCA is performed directly on the covariance matrix without standardizing. What is the most likely consequence?

    - ☐ Age will necessarily dominate because it has a smaller numerical range.

- ☐ Income may dominate the principal components because its scale produces much larger variance.
- ☐ PCA automatically normalizes all features before calculating covariance.
- ☐ Standardization has no effect on PCA because eigenvectors are scale-invariant.

12. Which statement correctly distinguishes bagging from boosting?

- ☐ Bagging trains models sequentially to focus on previous errors, while boosting trains them independently.
- ☐ Bagging primarily reduces variance through parallel models trained on resampled data, while boosting sequentially builds models that focus on reducing previous errors.
- ☐ Both methods necessarily train models sequentially using identical weighted datasets.
- ☐ Bagging primarily reduces bias, while boosting can only reduce variance.

13. Why can different initial centroid selections produce different K-Means results?

- ☐ K-Means always converges to the globally optimal SSE, but the number of iterations varies.
- ☐ K-Means minimizes a non-convex objective with respect to cluster assignments and centroids, so initialization can lead to different local minima.
- ☐ Centroids are selected randomly after every iteration.
- ☐ K-Means changes its distance metric depending on initialization.

14. A centered point is $x = (4, 2)$, and PCA's first principal component is

$$v = \frac{1}{\sqrt{2}}(1, 1).$$

What is its scalar projection onto the first principal component?

- ☐ 3
- ☐ $\sqrt{6}$
- ☐ $3\sqrt{2}$
- ☐ 6

15. In an AdaBoost round, $\varepsilon = 0.3$ which value is closest to α?

- ☐ 0.424
- ☐ 0.847
- ☐ 1.204
- ☐ 1.386

16. In a typical stacking ensemble, what is the primary purpose of the meta-learner?

- ☐ It replaces all base models with a single randomly selected model.
- ☐ It learns how to combine the predictions/features produced by base learners.
- ☐ It forces all base learners to use identical hyperparameters.
- ☐ It removes the need for a validation dataset.

17. Which situation is particularly suitable for DBSCAN rather than standard K-Means?

- ☐ Data consists of compact spherical clusters and the exact number of clusters is known.
- ☐ Data contains arbitrarily shaped dense clusters, noise points, and the number of clusters is unknown.
- ☐ Data contains only two numerical dimensions and no outliers.
- ☐ Data has categorical features and a predefined number of classes.

18. Which statement about a DBSCAN border point is correct?

- ☐ A border point must satisfy the MinPts requirement inside its own ε-neighborhood.
- ☐ A border point does not satisfy the core-point density requirement itself but lies within ε of a core point.
- ☐ A border point is always classified as noise.
- ☐ A border point must be farther than ε from every core point.

19. A cluster contains points $(1, 1)$, $(2, 2)$, $(3, 3)$ and centroid $(2, 2)$. What is the contribution of this cluster to K-Means SSE?

- ☐ 2
- ☐ 4
- ☐ $2\sqrt{2}$
- ☐ 8

20. Which statement most accurately describes a fundamental limitation of standard K-Means?

- ☐ It cannot work with numerical data.
- ☐ It assumes clusters are approximately spherical/convex under the chosen distance metric and requires K beforehand.
- ☐ It automatically determines the optimal number of clusters but cannot handle outliers.
- ☐ It requires labeled data to calculate cluster centroids.